

PPG-Based Rhythm Recognition for Wearable Health & Biometrics

EE3020A: Digital Signal Processing

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WHAT IS PPG?

Photoplethysmography (PPG) is an optical method used to measure blood volume changes.

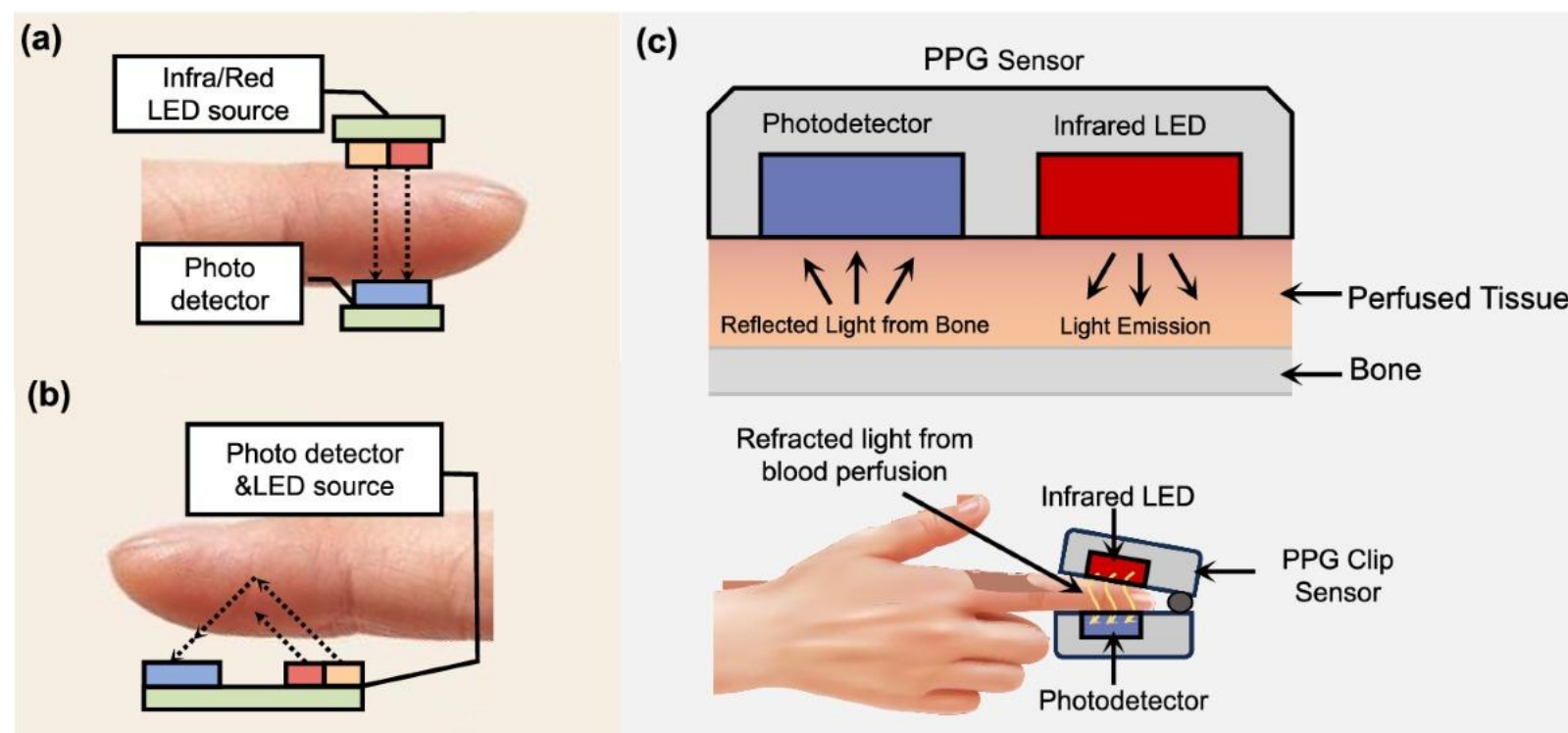
Motivation

Cardiovascular diseases are a major global health concern, and wearable PPG-based monitoring can enable early and continuous detection of abnormal heart rhythms.

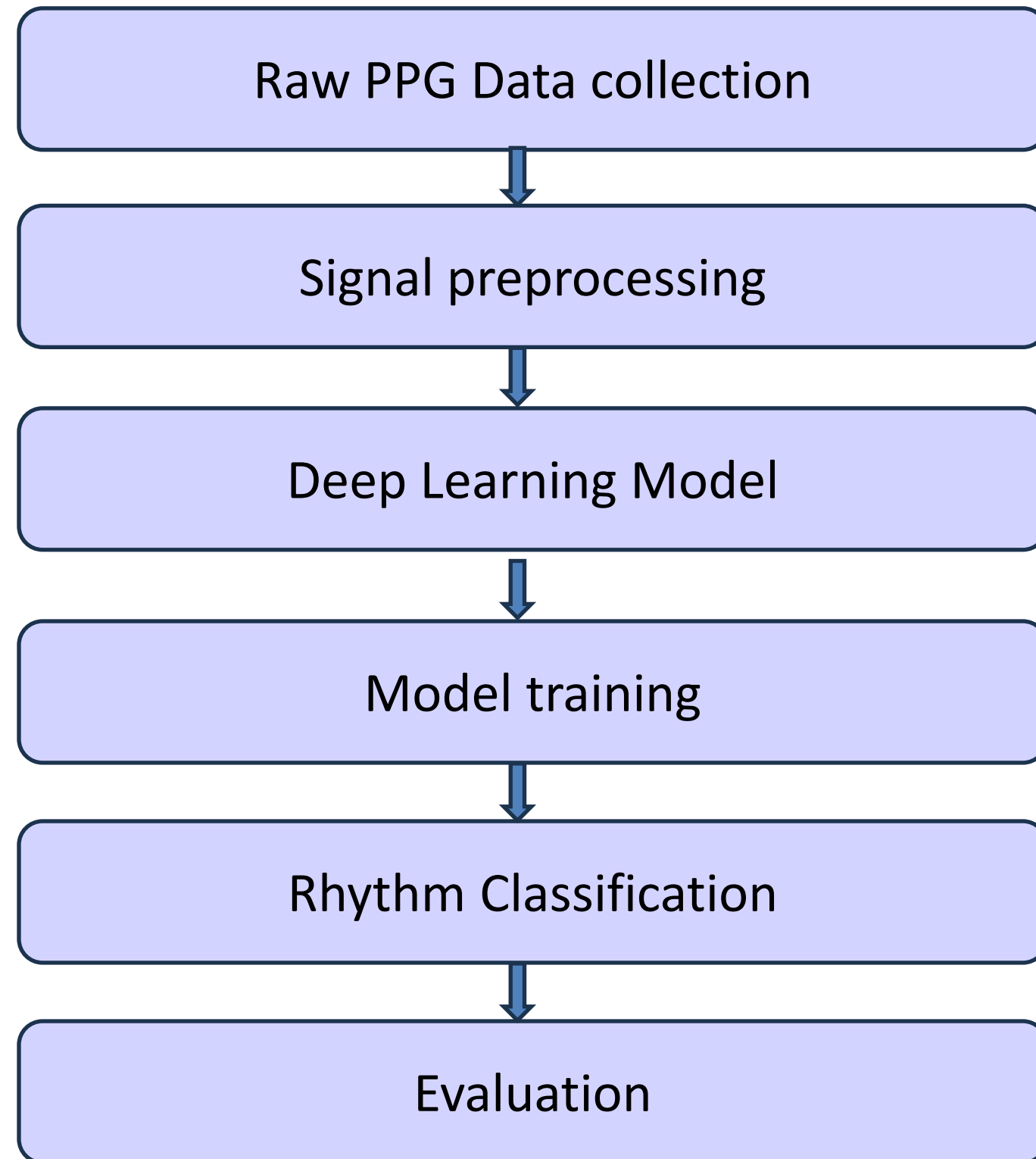
Continuous monitoring of heart rhythm is important for early detection of abnormalities.

Objective

To classify the ppg signal obtained from the wearable devices into regular or irregular by preprocessing and by using deep learning model specifically convolutional neural network model.



Methodology



DSP Formulas

Butterworth filter equation:

$$B_n(s) = \prod_{k=1}^{\frac{n}{2}} \left[s^2 - 2s \cos\left(\frac{2k+n-1}{2n} \pi\right) + 1 \right] \quad n = \text{even}$$

$$B_n(s) = (s+1) \prod_{k=1}^{\frac{n-1}{2}} \left[s^2 - 2s \cos\left(\frac{2k+n-1}{2n} \pi\right) + 1 \right] \quad n = \text{odd.}$$

Mean removal formula:

$$x_{\text{new}}(n) = x(n) - \mu$$

$x(n)$ = original signal/sample

μ = mean of the signal

$x_{\text{new}}(n)$ = mean-removed signal

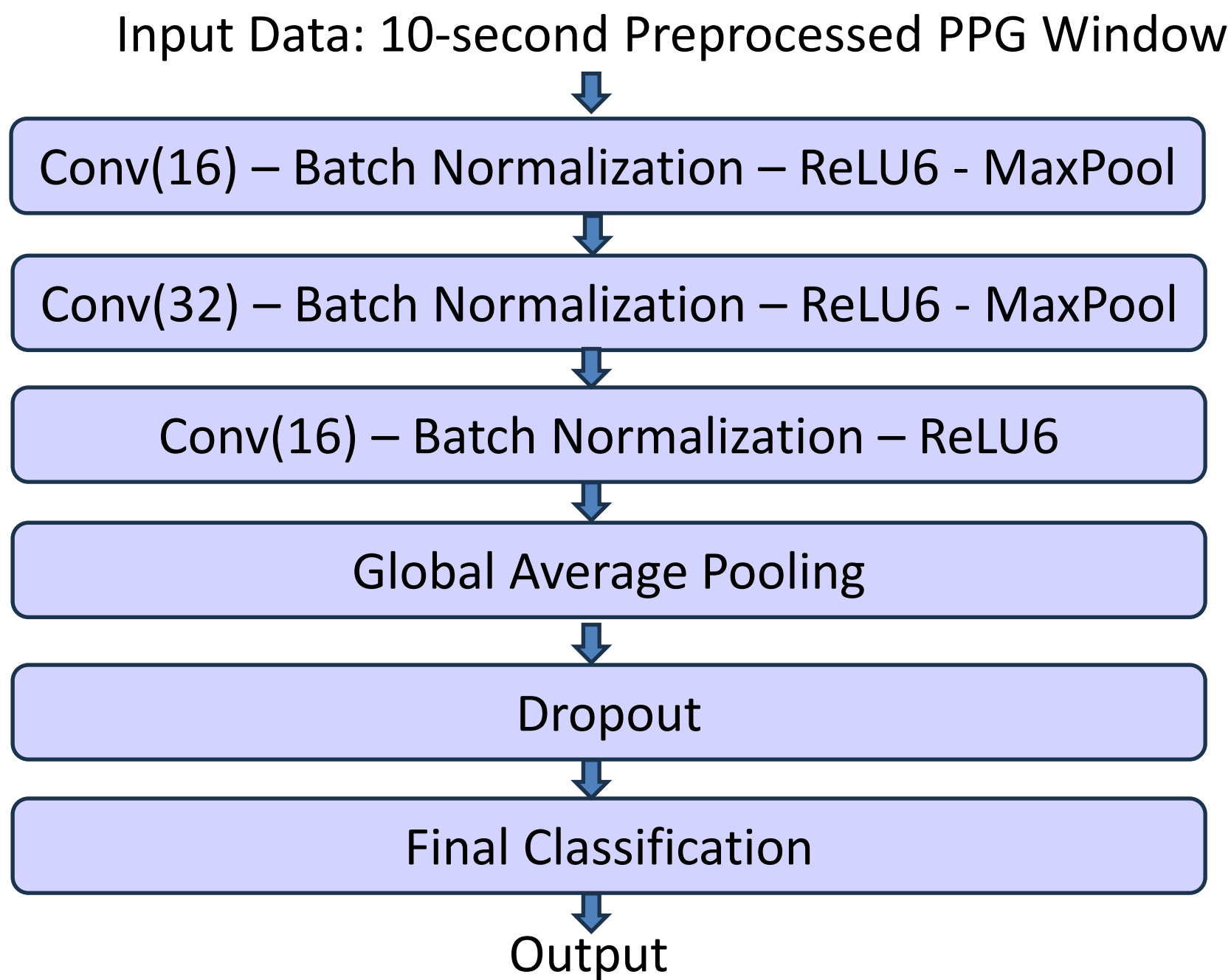
Normalization formula

$$z = \frac{x - \mu}{\sigma}$$

μ = mean

σ = standard deviation

Deep Learning Network Architecture

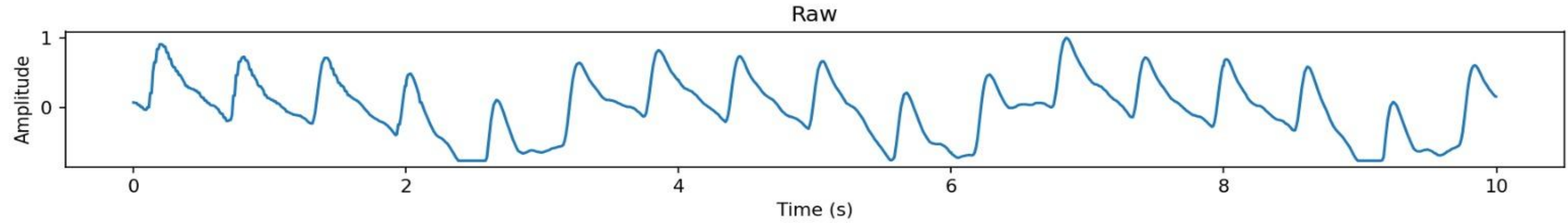


Final Hyperparameters

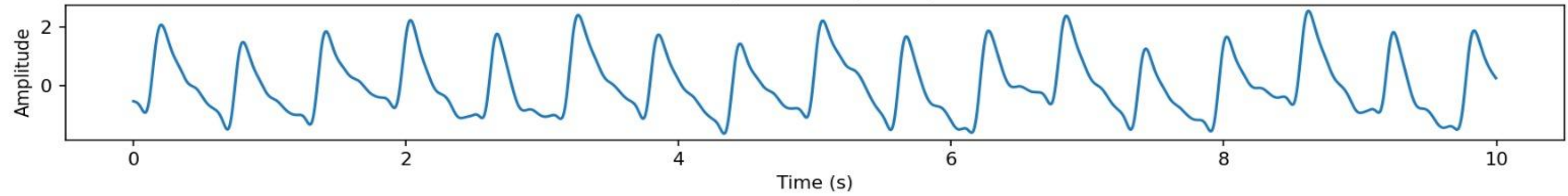
Number of layers:	3
Pool size	: 2
strides	: 1
padding	: valid
Dropout rate	: 0.1
Dense layer	: 1 (unit)
Activation	: sigmoid
optimizer	: Adam
Learning rate	: 0.0001
Loss function	: Binary cross entropy
Monitor	: Validation loss
Patience	: 15
Kernel size	: 3
Batch Size	: 32

Waveforms

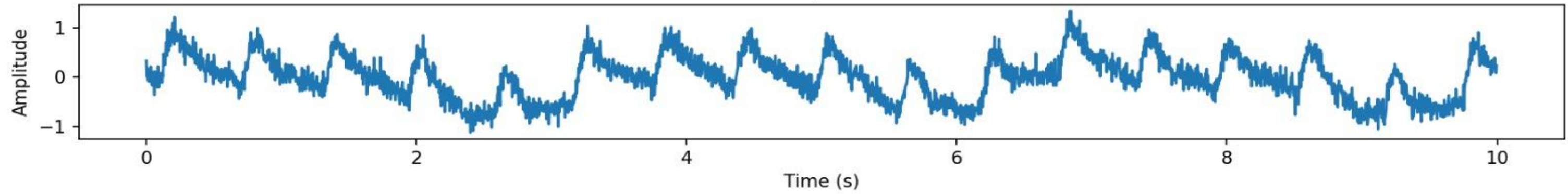
Regular Signal (10 sec)



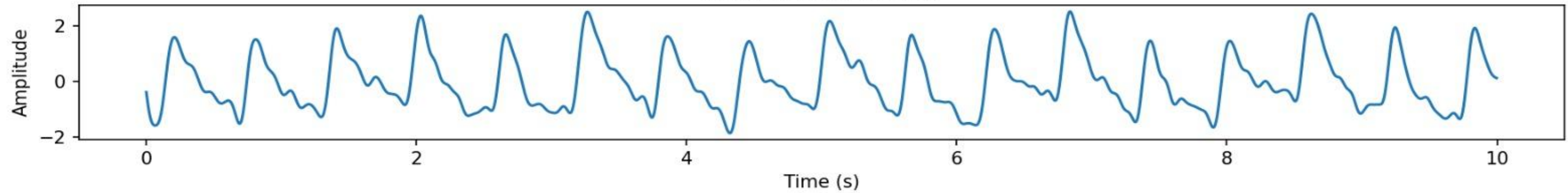
Preprocessed (Clean)



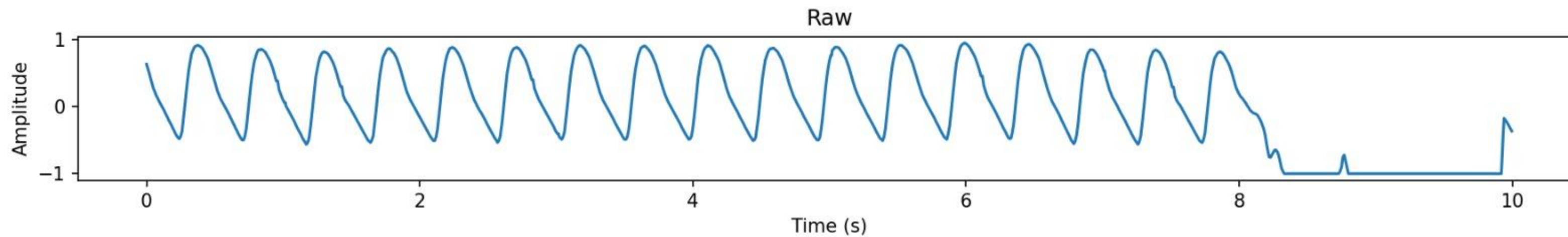
Noisy Raw



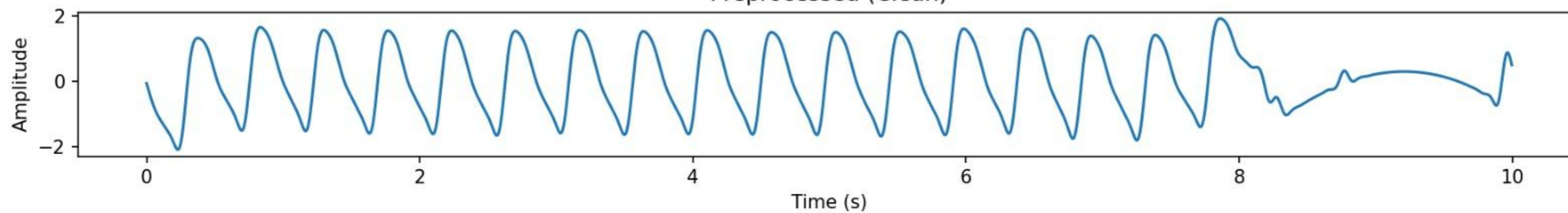
Preprocessed (Noisy)



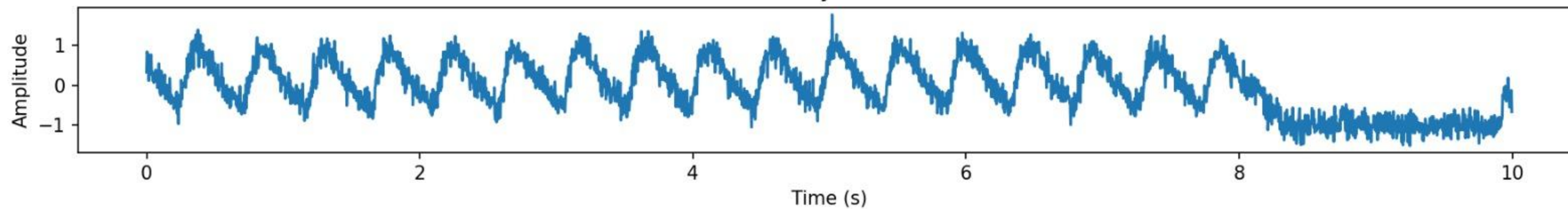
Irregular Signal (10 sec)



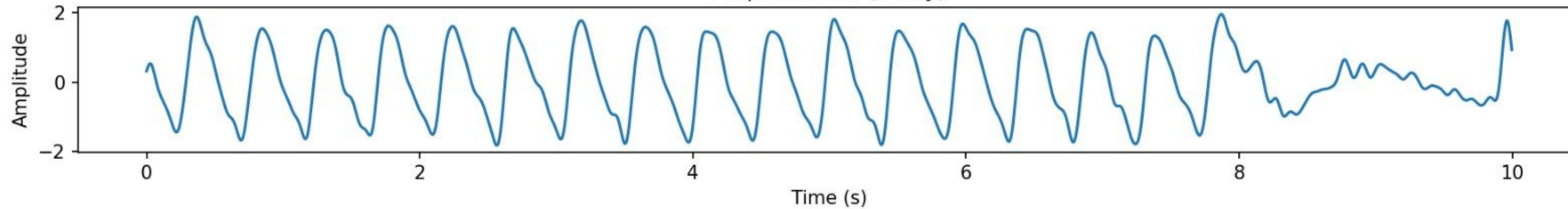
Preprocessed (Clean)



Noisy Raw



Preprocessed (Noisy)





Dataset Description

Source: CapnoBase PPG (Fs = 300 Hz, 10-sec windows)

Preprocessing: 3rd-order Butterworth Bandpass (0.5–10 Hz)

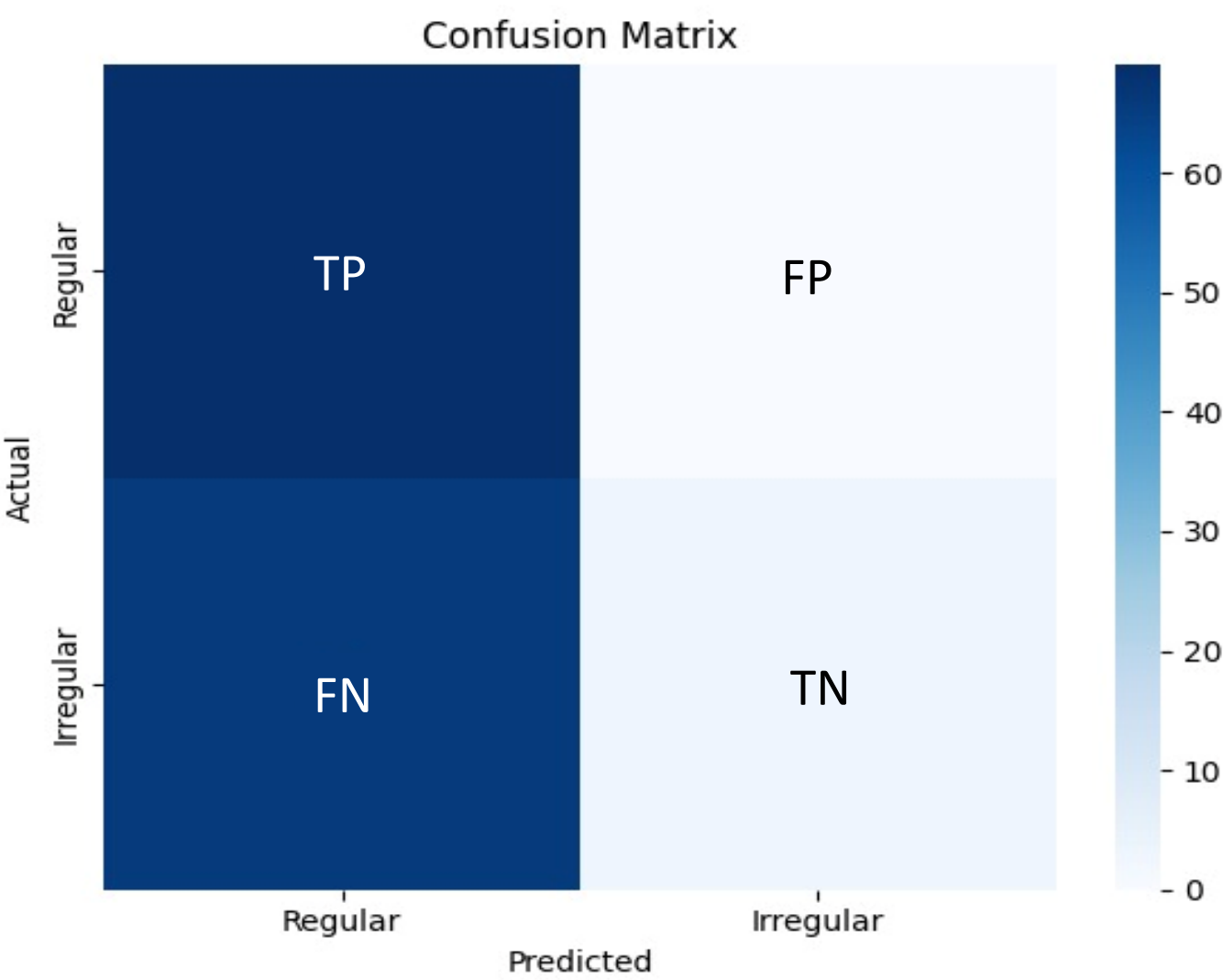
DATASET DESCRIPTION:
Regular: 1000, Irregular: 1000

Training data: 1600 (800+800)
Validation data: 200 (100+100)
Testing data: 200 (100+100)

Evaluation Metrics

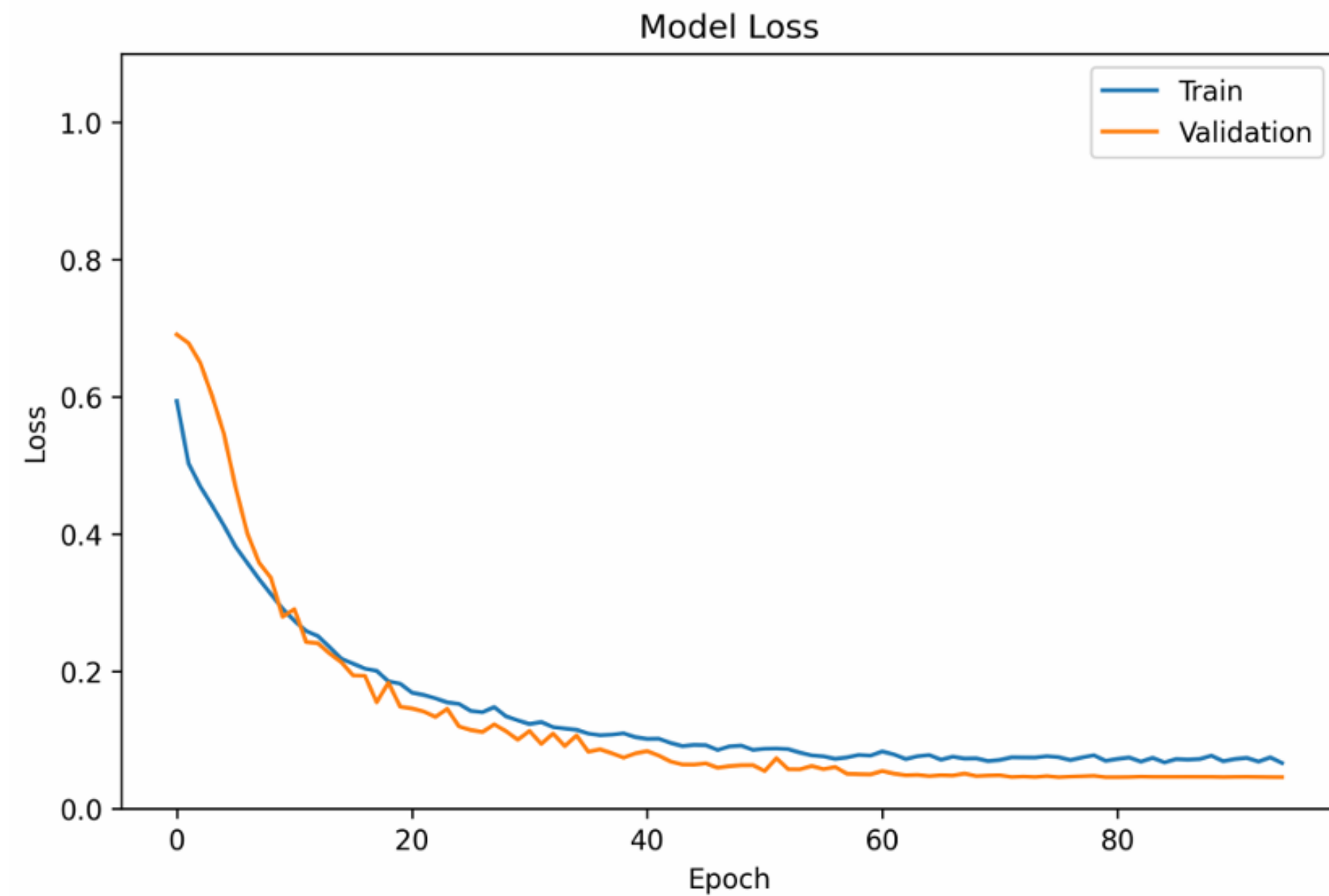
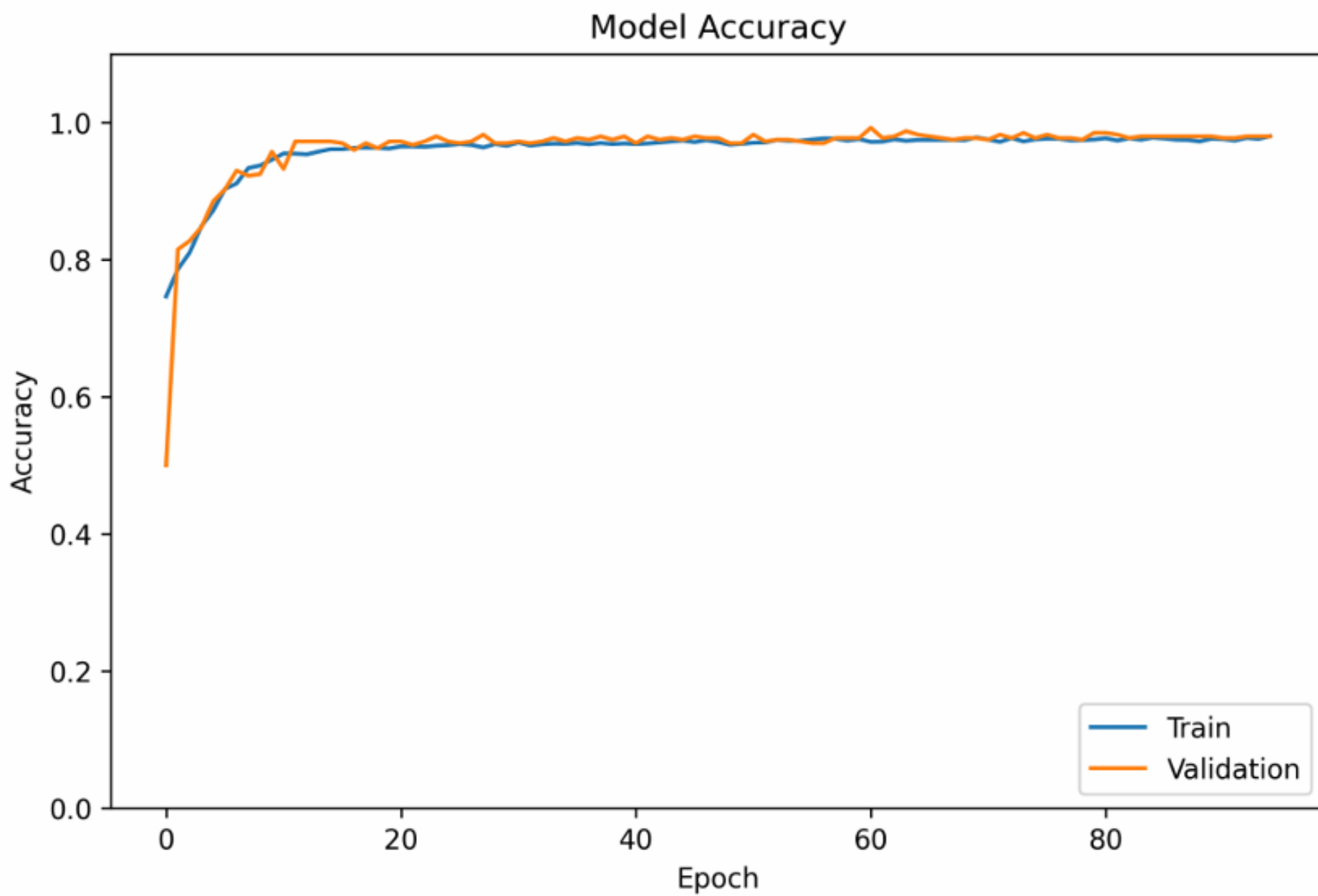
Accuracy		Precision
$\frac{TP + TN}{TP + TN + FP + FN}$		$\frac{TP}{TP + FP}$
	F1 Score	
	$2 \cdot \frac{\text{Precision} \cdot \text{Sensitivity}}{\text{Precision} + \text{Sensitivity}}$	
Sensitivity		Specificity
$\frac{TP}{TP + FN}$		$\frac{TN}{TN + FP}$

Results:



Activation Function	Confusion Matrix				Performance Metrics			Testing time(ms)	Model size(MB)
	TP	FP	FN	TN	Train Accuracy (%)	Valid Accuracy (%)	Test Accuracy (%)		
ReLU	200	0	8	192	98.03	98	98	195.2	
ELU	200	0	23	117	92.81	94.50	94.25	223.86	0.108
ReLU6	200	0	12	188	97	97.75	97	223.861	0.108

Training Curves



Observations



From DSP Techniques:

Butterworth filter was chosen to apply because it has maximally passband. Since for PPG signal CNN relies on the shape and timing of the pulse to detect irregular rhythms, there should not be any ripples in the passband.

The lower cutoff frequency of the bandpass filter is taken to be 0.5Hz so that it eliminates respiratory artifacts in the signals.

The higher cutoff frequency of the bandpass filter is chosen to be 10Hz to ensure that it eliminates the powerline interference and sharp noise artifacts.

Normalization is done so that CNN model focusses on the shape rather than the amplitude.

Preprocessing helped the model to learn the required features from the shape to classify the heart rhythms, without concentrating on remembering the noises.

From Deep Learning Techniques:

Global average pooling kept the final model size under 110KB proving its viability for smart watch hardware.

It was observed that kernel size of 3 enabled the network to focuss mainly on pulse shape and peak-to-peak timing features rather than the amplitude of the signal.

Another important observation is that implementation of early stopping and dynamic learning rate adjustment prevented noticeable overfitting during training.

Confusion matrix analysis showed that there is high sensitivity in detecting abnormalities while maintaining less false alarm rates for normal rhythms.



Thank You :)

